## 1 Label-free graph representation

Input graph
$G = (V, E, X)$

Node features X

Adjacency A

| | $v_1$ | $v_2$ | ⋯ | $v_n$ |
|---|---|---|---|---|
| $v_1$ | 0 | 1 | ⋯ | 0 |
| $v_2$ | 1 | 0 | ⋯ | 1 |
| ⋯ | ⋮ | ⋮ | ⋱ | ⋮ |
| $v_n$ | 0 | 1 | ⋯ | 0 |

Two augmented views

edge dropping + feature masking

independent augmentation

**Shared GRACE encoder**
$g_\psi$

Embedding $H^1$

Embedding $H^2$

Frozen normalized representation $\bar{H}$

## 2 Label-agnostic granular-ball partition

Geometry-only construction in frozen embedding space $\bar{H}$

Initialization
$B_0 = V$

Recursive bi-partition

Further recursive splitting

Embedding space $\bar{H}$ covered by granular balls $B = \{B_j\}_{j=1}^{B}$

selected ball $B_j$

Label-agnostic recursive 2-means; projected ball covers are shown for visualization

## 3 Leave-one-out ball-level label consensus

(a) Select a granular ball $B_j$

(b) Leave-one-out consensus

$p_{j,-v}$ (exclude v)

1 2 3 4 5

$$q_{j,-v} = (1 - \beta)p_{j,-v} + \beta e_m$$

If the majority ties: $q_{j,-v} = p_{j,-v}$
(tie-safe; no arbitrary one-hot class)

(c) Construct a soft target $t_v$

noisy one-hot

local consensus

1 2 3 4 5

$$t_v = (1 - \alpha_{j,-v})e_{\tilde{y}_v} + \alpha_{j,-v}q_{j,-v}$$

$\alpha_{j,-v}$ measures vote concentration
$\beta = 0.5$ in the fixed configuration

## 4 X-only MLP training and inference

Training with graph-derived targets

node feature $x_v$

2-layer MLP $f_\theta$

precomputed $t_v$

**Soft cross-entropy** prediction vs. $t_v$

Inference after training (transductive)

trained MLP

node feature x

class probabilities
**no graph access**

Class 1 Class 2 Class 3 Class 4 Class 5

*Class colors denote the 20% training nodes with observed noisy labels; unlabeled nodes are omitted.*

data flow

contrastive alignment

**X-only inference:** no A, no $\bar{H}$, no neighbors